

WEEK 10 – FINAL DOCUMENTATIO

Scientific Calculator

JavaSwingProject

Student: Md Omor

Faruk Student ID:

11240321755

1. Introduction

This document presents the final documentation for the Scientific Calculator project developed using Java Swing. The purpose of this documentation is to summarize the final system, its major features, testing results, screenshots to be included, and final submission preparation completed during Week 10.

2. Project Overview

The Scientific Calculator is a GUI-based Java application designed to perform basic arithmetic and scientific calculations. The application provides an easy-to-use calculator interface with ON/OFF control, display, history/status information, arithmetic operators, scientific functions, error handling, and keyboard input.

3. Final System Features

- ON/OFF power control.
- Basic arithmetic operations: addition, subtraction, multiplication, and division. Power operation using x^y .
- Scientific functions: sin, cos, tan, log, ln, x^2 , $\sqrt{}$, $1/x$, and percentage.
- Constants: π and e .
- ANS function for recalling the previous result.
- Positive/negative number control using $\pm$.
- Decimal number input.
- AC, CE, and Backspace controls.
- Direct expression input such as sin(90), cos(90), tan(45), and log(100).
- Keyboard input support for direct expressions.
- Error handling for division by zero and invalid mathematical domains.
- History and status display for calculation feedback.

4. User Interface

The final calculator interface uses a fixed window size of 440×680 pixels. The original GUI structure, colors, button layout, fonts, and overall calculator appearance are maintained in the final version.

5. Final Testing

The following test cases were used to verify the final system:

Test Input	Expected Result	Purpose
10 ÷ 0	Div/0	Division-by-zero error
√(-9)	Domain Domain	Invalid square-root input
log(0)	undef	Invalid logarithm input
tan(90)	0	Undefined tangent Scientific
sin(180)	0	function accuracy Scientific
cos(90)	1	function accuracy Scientific
sin(90)	1	function accuracy Scientific
cos(0)	1	function accuracy Scientific
tan(45)	2	function accuracy Logarithm
log(100)	5	calculation Square-root
√(25)	25	calculation Square
5 ²		calculation

6. Direct Expression Testing

The final version supports direct scientific expressions. The user can enter an expression such as sin(90) using the keyboard and press the = key or Enter key to obtain the result.

Expression	Result
sin(90)	1
cos(90)	0
tan(45)	1
log(100)	2
ln(10)	2.302585093
√(25)	5

7. Screenshots

The following screenshots should be inserted in the final report to document the completed system:

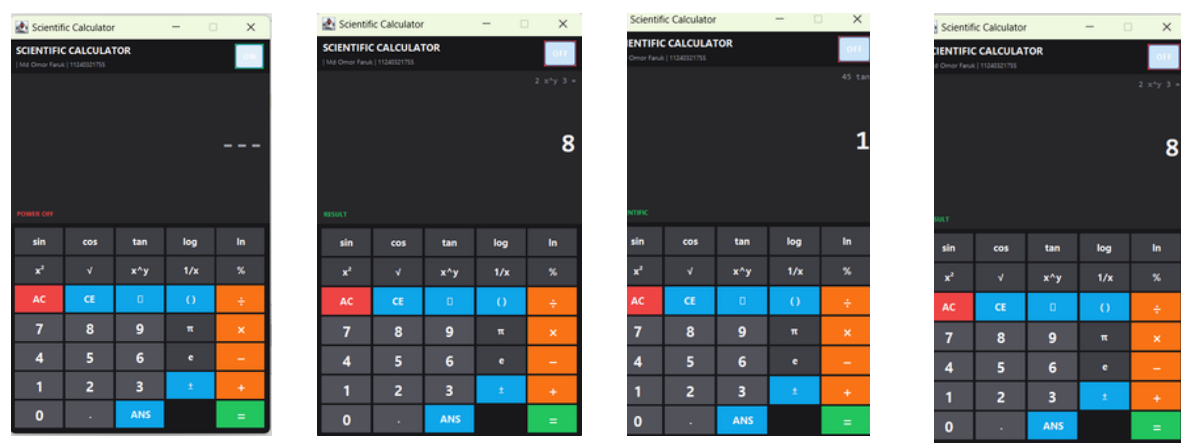


Figure 1: Calculator interface in ON/READY state.

Figure 2: simple Addition

Figure 3: Result of tan(45) showing 1.

Figure 4: Example of a basic arithmetic calculation.

8. Documentation and Submission Preparation

- Final Java source code is organized and ready for submission. Final GUI screenshots are prepared for the report.
- Testing results are documented.
- Project features and system behavior are described.
- Final report and supporting files are organized for submission.

9. Conclusion

The final documentation summarizes the completed Scientific Calculator project and its testing process. The final system provides both basic and scientific calculation features, direct expression input, keyboard support, and error handling. The documentation and screenshots can be included with the final project files for submission.

10. References

- Java Platform, Standard Edition (Java SE) documentation.
- Java Swing GUI component documentation.
- Project source code and weekly work-plan materials.

